Universidade de Brasília
Faculdade do Gama

Digital Integrated Circuit Design

Laboratory 1 – MOS Transistor Simulation

Objective:

To visualize the **Id** curve as a function of $\mathbf{V_{ds}}$ for **NMOS** and **PMOS** transistors.

Schematic:

Create your personal folder on the **Desktop**. Inside your personal folder, create a new subfolder named **Lab1**. Then, open a terminal within the **Lab1** directory and launch **Cadence Virtuoso** by entering the following commands:

```
zsh
virtuoso
```

First, we need to create a working library. Open the **Library Manager** window and navigate to **File** → **New** → **Library**. Create a new library named **curves** inside your **Lab1** folder. In the **Technology File for New Library** window, select the option **Attach to an existing technology library**, and then click **OK**. In the following window, choose the **gpdk045** library.

Now, we need to create a schematic. In the **Library Manager,** select the newly created library and go to **File** → **New** → **Cell View**. In the **New File** window, type **nmos_curves** in the **Cell** field and click **OK**. The schematic editor will open.

From the schematic editor, go to **Create** → **Instance** to add an NMOS transistor. In the **Add Instance** window, click **Browse** and configure the required fields as follows:

Library: gpdk045
Cell: nmos1v
View: symbol

After configuring the instance, click **Hide** and place the NMOS transistor in the schematic editor.

Similarly, instantiate two **vdc** voltage sources from the **analogLib** library. For the first source, set the **DC Voltage** field to **Vg**. For the second source, leave the **DC Voltage** field blank. Finally, instantiate the ground (**gnd**).

Go to **Create** → **Wire (narrow)** and connect the components as shown in **Figure 1**. Once all connections are complete, select **Check and Save** to verify and save the schematic.

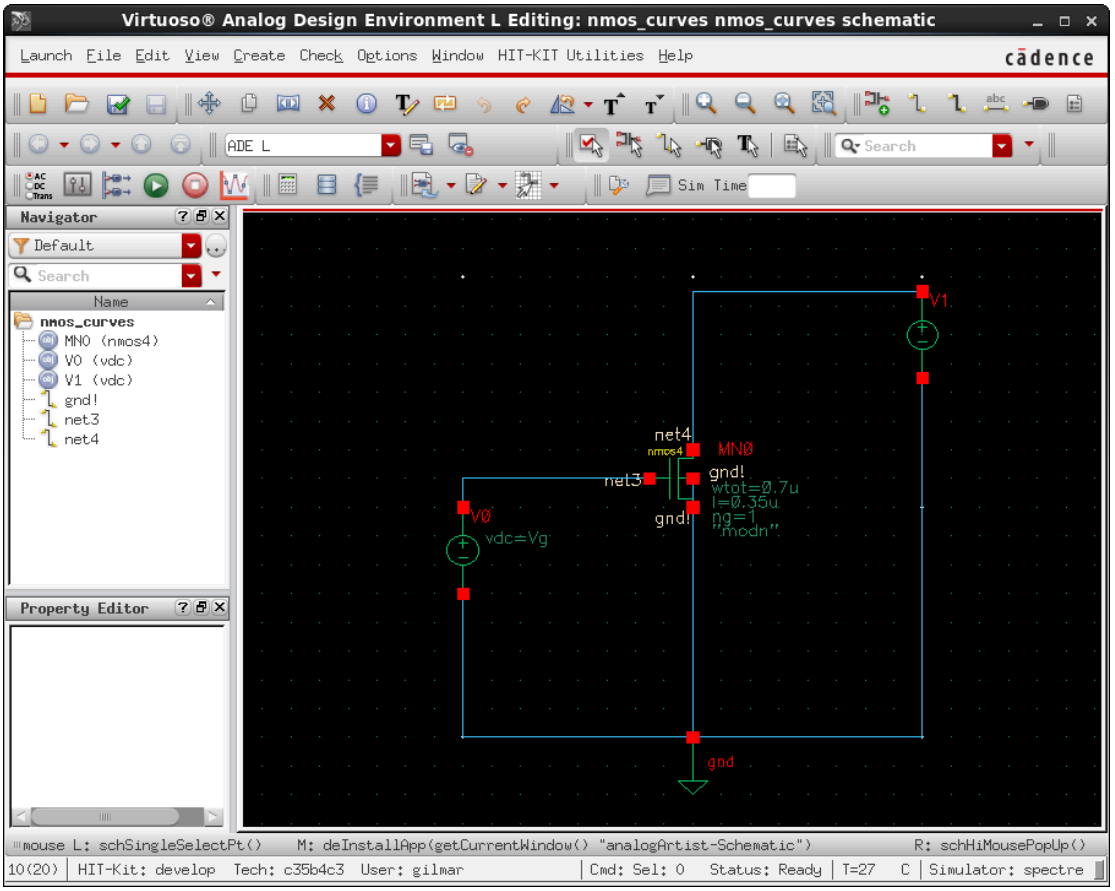


Figure 1

Simulation:

In the schematic editor window, go to **Launch → ADE Explorer** and select **Create New View**. Verify the information in the fields as shown in **Figure 2**. Choose whether to open the **maestro view** in a **new tab**, a **new window**, or in the **current tab**, according to your preference.

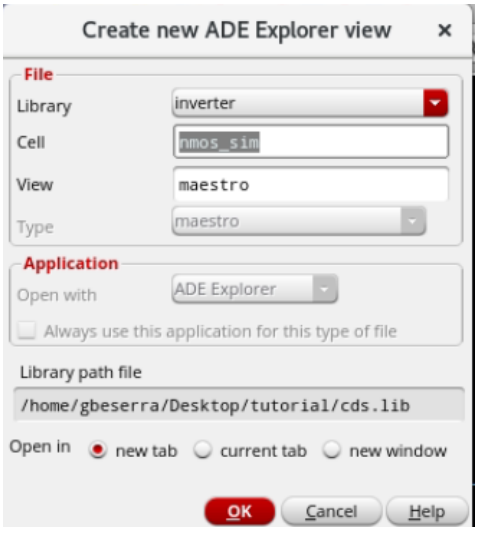


Figure 2

In the **ADE Explorer** window, navigate to **Analysis → Choose** and select the **dc** option. Under **Sweep Variable**, choose **Component Parameter**. Click **Select Component**, then select the voltage source **V1** in the schematic and specify the **dc** parameter (see **Figure 3**).

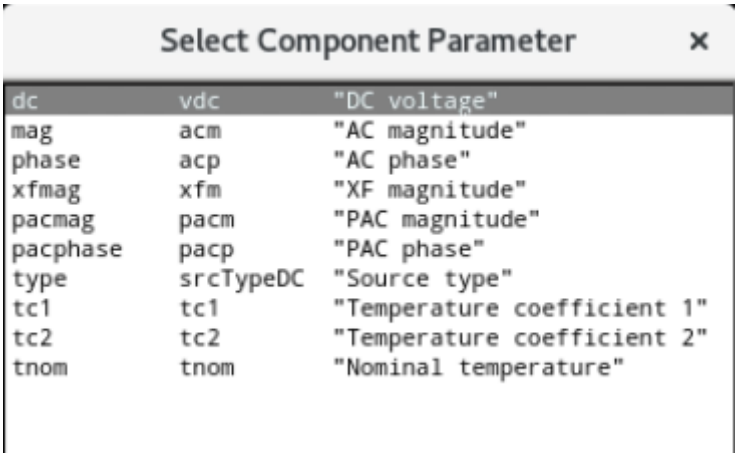


Figure 3

Return to the **Choosing Analyses** window and complete the **Sweep Range** fields with the desired values. After entering the following parameters, click **OK** (see **Figure 4**).

Start: 0
Stop: 1.1
Sweep Type: Linear
Step Size: 0.1

In the **maestro** tab, right-click on **Design Variables** and select **Copy from Cellview**. The **Vg** variable should appear in the list. In the blank field next to **Vg**, click on the **three dots** to open the **Parameterize** window. From the **Add Specification** menu, choose **From/To** and fill in the fields as shown in **Figure 5**.

In the **Outputs** menu, select **To Be Plotted → Select on Design**. Click on the **negative terminal** of the **V1 source** in the schematic (a red square), as shown in **Figure 6**. The **/V1/MINUS** terminal will appear on the **Outputs Setup** tab of the **maestro** window (see **Figure 7**). This signal corresponds to the NMOS drain current (**Id**).

Finally, go to **Simulation → Netlist and Run**, or simply click the **green arrow icon**, to execute the simulation. Check the resulting waveforms displayed in the output window.

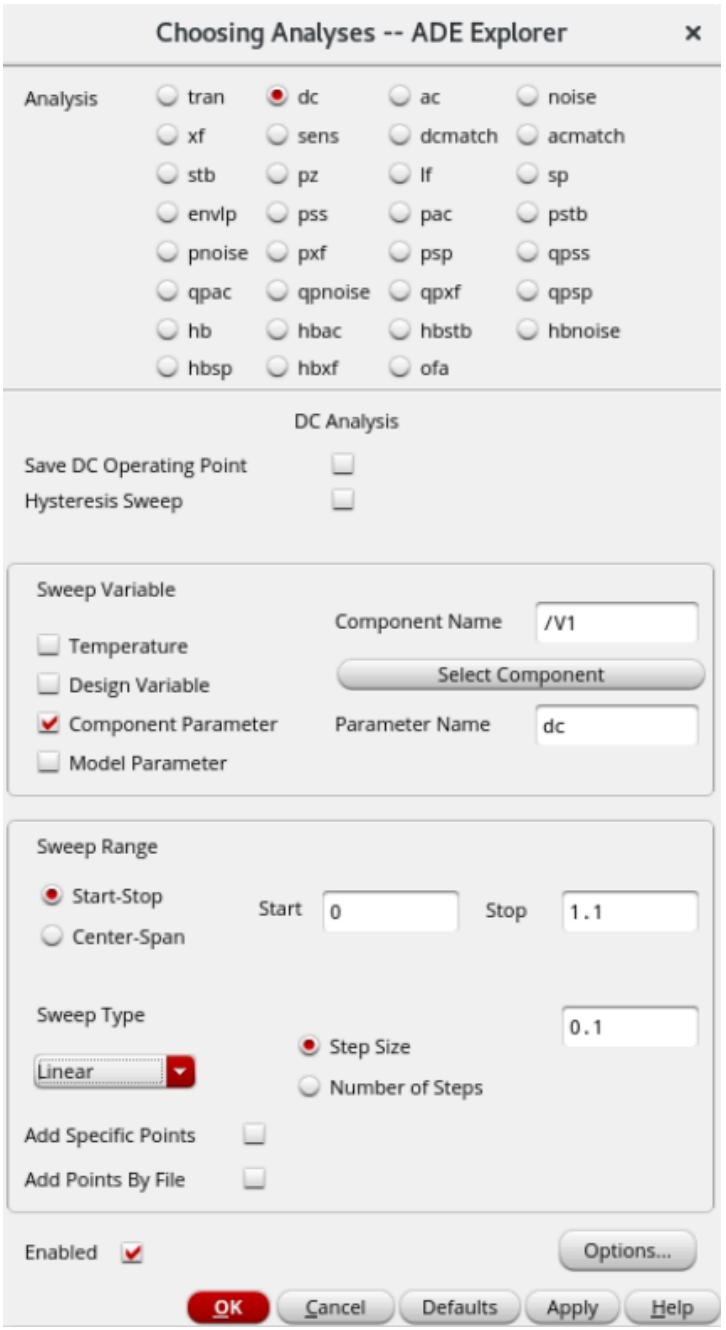

Choosing Analyses -- ADE Explorer
Analysis
tran
dc
ac
noise
xf
sens
dcmatch
acmatch
stb
pz
lf
sp
envlp
pss
pac
pstb
pnoise
pxf
psp
qpss
qpac
qpnoise
qpxf
qpsp
hb
hbac
hbstb
hbnoise
hbsp
hbxf
ofa
DC Analysis
Save DC Operating Point
Hysteresis Sweep
Sweep Variable
Component Name
/V1
Temperature
Select Component
Design Variable
Component Parameter
Parameter Name
dc
Model Parameter
Sweep Range
Start-Stop
Start
0
Stop
1.1
Center-Span
Sweep Type
0.1
Step Size
Linear
Number of Steps
Add Specific Points
Add Points By File
Enabled
Options...
OK
Cancel
Defaults
Apply
Help

Figure 4

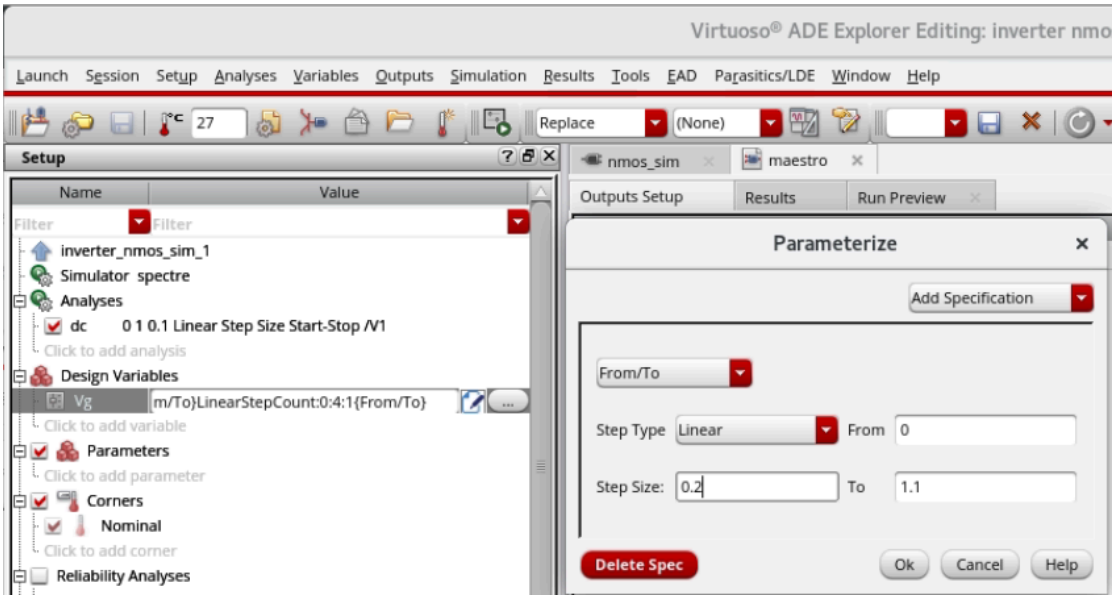


Figure 5

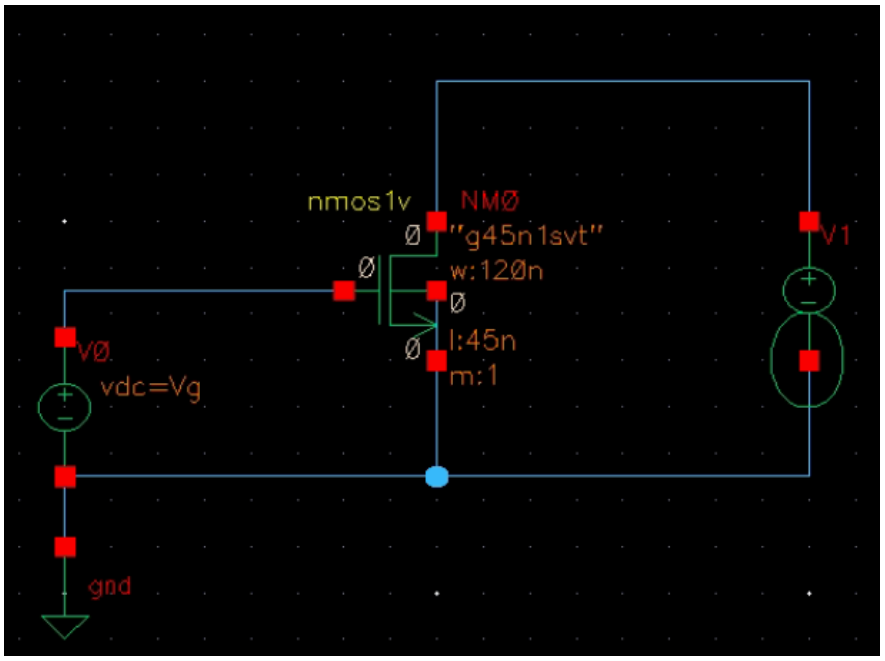


Figure 6

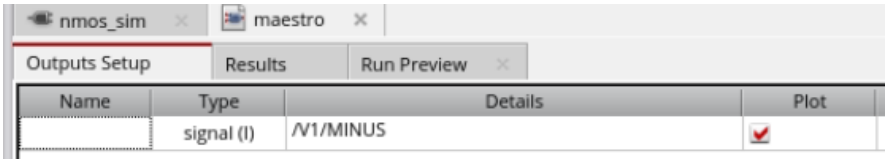


Figure 7

Task:

Submit a PDF file containing:

a) A brief description of the purpose of the simulation performed.
b) A screenshot of the waveforms, identifying the cutoff, linear, and saturation regions of the NMOS transistor.
c) A screenshot of the same waveforms for a PMOS transistor.
Hint: Use the schematic shown in Figure. 2 to perform this simulation.

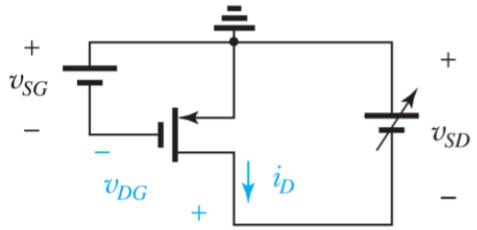


Figure 2

d) A brief explanation of the differences between NMOS and PMOS transistors.